09 · Does the "€100 -> Gold tier" perk retain people? — RDD (CausalPy)

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> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The decision. Customers who cross **€100** annual spend get **Gold** status and its perks. Gold members retain better — but of course they do, they're big spenders. Does the *perk itself* boost retention, or are we just seeing that big spenders retain anyway? And where should we set the threshold?

Regression discontinuity exploits the sharp €100 rule: customers *just below* and *just above* €100 are essentially identical except for the perk, so a **jump** in retention exactly at the cutoff is caused by the perk. We estimate it, then run the full RDD validity battery — **McCrary** density (manipulation), **bandwidth** sensitivity, **placebo cutoffs**, and **polynomial order** — because a headline RD number without those checks is not trustworthy.

7-step contract.

1.1 2 · Simulate a ground truth

Retention rises smoothly with annual spend (big spenders retain more anyway) — **plus** a true **+14pp jump** exactly at €100 from the Gold perk. The smooth part is the confounder; the jump is the causal effect. Spend is concentrated around the cutoff so both sides are well-populated.

```
TRUE perk effect = +14% retention at €100  ·  50% are Gold
```

	spend	treated	retention
0	84.325206	0	1.0
1	101.996816	1	0.0
2	93.392393	0	1.0
3	93.281224	0	0.0
4	102.504635	1	0.0

1.2 3 · Identify — a jump at the cutoff is causal (and its validity conditions)

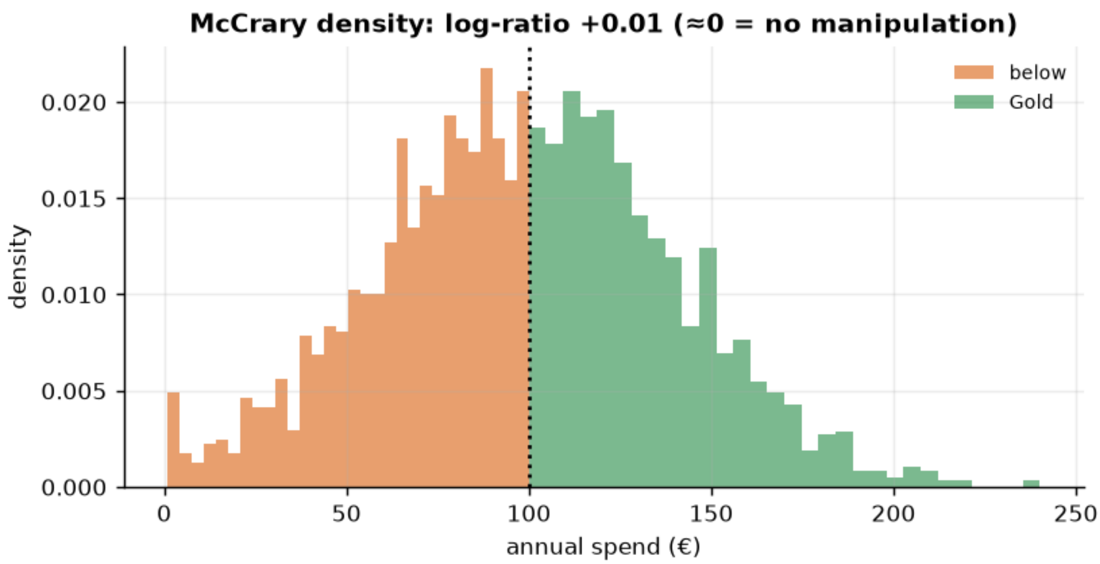
Sharp RD estimand: $\tau_{RD} = \lim_{x\downarrow c} \mathbb{E}[Y \mid X{=}x] - \lim_{x\uparrow c} \mathbb{E}[Y \mid X{=}x]$ — the discontinuity in retention at spend = €100. **Identification:** potential-outcome means are *continuous* at the cutoff, so units just either side are comparable and the jump is causal. This is a **LATE at the cutoff** — local to €100. Two things must hold, both checked below:

- **No manipulation** of the running variable at the cutoff (customers/business can't sort across it) — the McCrary density test.
- **The functional form isn't driving the jump** — checked via bandwidth, placebo cutoffs, and polynomial order.

1.3 3b · Validity check 1 — McCrary density (manipulation)

If customers game their spend to just clear €100 (or the business nudges them), the *density* of spend jumps at the cutoff and RD breaks. We compare the density just below vs just above.

```
density below 23.7 vs above 24.0, log-ratio +0.01 - no sign of sorting across
the cutoff.
```



1.4 4 · Estimate — Bayesian regression discontinuity

Local-linear fit on each side within a **bandwidth** of €40 (the comparable customers). Wider borrows non-comparable whales; narrower throws away data.

```
RD jump +11.3% (true +14%)  ·  90% CI [+2.8%, +19.2%]
```

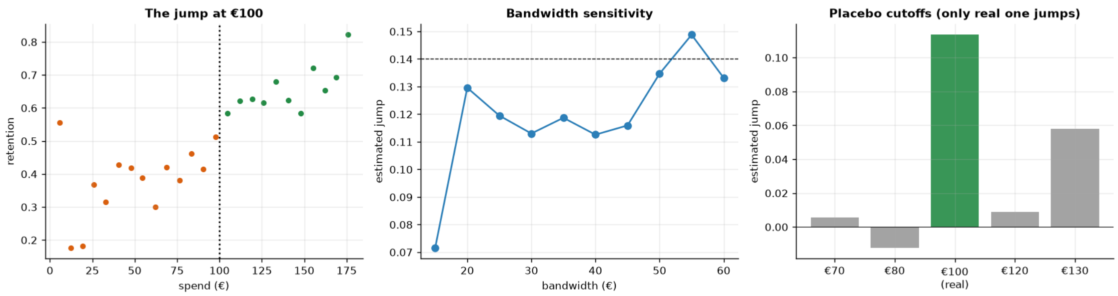
1.5 5 · Validate — see the jump, plus bandwidth / placebo / polynomial robustness

A credible RD number is one that **survives** the researcher-degrees-of-freedom checks:

- **Bandwidth curve** — the estimate should be stable across a sensible range of bandwidths.

- **Placebo cutoffs** — run RD at fake thresholds (€70, €130) where there's no perk; the jump should be ~ 0. A real jump at a fake cutoff would mean the method finds discontinuities in noise.
- **Polynomial order** — linear vs quadratic local fit shouldn't move the answer much.

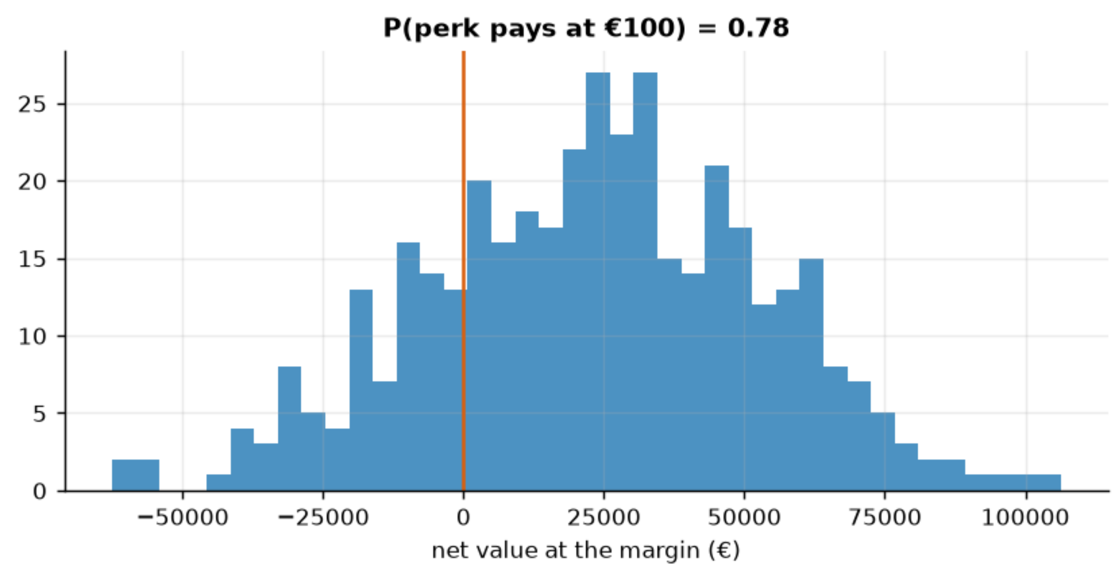
```
Bandwidth-stable around +12.0%.  Polynomial: linear +11.3%, quadratic +10.9%.
Placebo cutoffs: ['+0.6%', '+5.8%'] (~0).
```



1.6 6 · Decide, in euros — does the perk pay, and where to set the cutoff?

Value the retention lift: extra retained customers x annual value, minus the perk cost, **at the current €100 margin** (that's what RD identifies). We also sweep the perk cost to find the break-even.

```
Net €7.7/member  ·  total €23,071 [90% €-28,191, €70,074]
P(perk pays) 0.78 -> reconsider  ·  break-even perk cost ~ €6/member
(conservative).
```



1.7 7 · Caveats

- **RD is *local.*** The effect is identified only *at the cutoff* — the perk's effect on customers near €100, not on your €500 whales. Don't extrapolate the jump to everyone.
- **Manipulation breaks it.** We checked the McCrary density; watch for heaping at round numbers.
- **Bandwidth is bias–variance.** We showed the estimate is stable across bandwidths; a fragile estimate that swings with bandwidth is a red flag.
- **Fuzzy RD if the rule isn't sharp.** If crossing €100 only *raises the probability* of Gold (grandfathering, overrides), use the fuzzy-RD ratio (jump in Y ÷ jump in P(treated)).